

QUESTION-TO-SLIDE MAPPING

SECTION A - LECTURE 04

Pricing & Bidding

Course	BECM 3101 - Construction and Project Management I
Lecture deck	4.0 - Pricing & Bidding (17 PDF pages)
Question bank	Term-final examinations, 2016-2024 (25 pages)
Method	Every relevant question is mapped to its primary and supporting lecture slides, regardless of the original Section A/B label.

RESULT

This is a high-yield lecture with relevant questions in every examination year. The strongest recurring blocks are bid unbalancing/front payment, contractor-side project development, competitive-bidding probability and the tender-price structure.

Prepared as part of the cumulative lecture-by-lecture question-bank analysis.

1. Executive Summary

The lecture contains tendering procedure, contractor-side project development, basic pricing equations, bid unbalancing/front payment, competitor-based bidding models and the Alpha historical-bid exercise. Slide numbers below use the PDF page count, with the title page counted as Slide 1.

17	25	13	3	9	286
Slides reviewed	Question links	Exact matches	Direct matches	Partial matches	Mapped marks

Classification	Meaning
Exact	The requested content, structure or numerical exercise appears essentially as asked in the lecture.
Direct	The lecture supplies the main answer and method, but the exam changes framing, data or required depth.
Partial	Only part of the question or the base formula/context is present; another lecture or class note is needed.

Slide-numbering note	PDF Slide 4 is printed as slide number 3 in the deck, because the title page is unnumbered. This report always uses the PDF page/slide number shown by the file viewer.
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Highest-priority slide blocks

Priority	Slides	Topic	Why it matters
1	8-9	Bid unbalancing and front payment	7 exam appearances, including the 2024 scenario-based question; the most repeated topic in this deck.
2	5-6	Contractor-side project development process	5 appearances from 2016-2022; usually requested with a diagram.
3	12-16	Competitor markup, histogram and probability curve	3 long numerical/graph questions; Slide 16 reproduces the Alpha dataset.
4	7	Tender-price and markup structure	Supports builder-cost structure, average-bid-price questions and the 2024 comparison problem.
5	10-11	Bidding uncertainty and model practicality	Useful supporting theory, especially for zero-profit-bid rationale, but not a complete answer by itself.

2. Detailed Year-wise Question Mapping

Question wording has been lightly standardized for readability without changing the assessed meaning. 'Primary' identifies the best slide(s) to begin from; 'Support' lists additional slides. 'QB p.' is the page in the uploaded question bank.

Examinations 2016-2018

Year	Original	Marks	QB p.	Question / task	Match	Primary	Support	Coverage assessment
2016	Sec A 1(b)	8	1	Explain the construction project development process with a diagram.	Direct	5-6	4	Slides 5-6 provide the contractor-side development flow from advertisement and bid study through planning, execution, control and completion. The exam wording is broader than the deck's contractor perspective.
2017	Sec B 7(a)	7	5	From the builder's perspective, write down the basic structure of construction cost.	Exact	7	5	Slide 7 gives the core tender-price equation and a numerical example; Slide 5 reinforces $\text{Price} = \text{Cost} + \text{Profit}$ and shows cost, overhead and markup components.
2017	Sec B 7(b)	3	5	List the steps for a typical tendering procedure.	Exact	4	-	Slide 4 presents the full nine-stage sequence from tender-document preparation to contract signing.
2017	Sec B 7(d)	15	6	Using Alpha Private Limited's 62 historical bids, draw a histogram, convert it to a probability curve and find the chance of winning at 7.5% markup.	Exact	12-16	10-11	Slide 16 reproduces the same data table and target markup. Slides 12-15 explain competitor identification, markup computation, histogram construction and probability conversion.
2017	Sec B 8(a)	5	6	A well-known contractor decides to submit a bid with zero profit. Critically explain the rationale.	Partial	10-11	5, 7	The deck discusses competitive uncertainty, quality weightage and bidding models as decision-support tools, but it does not provide a dedicated checklist of reasons for zero-profit bidding.
2018	Sec A 3(b)	15	7	Describe bid unbalancing and upfront/front-payment methods as parts of a bidding strategy.	Exact	8-9	-	Slides 8 and 9 are devoted to these two strategies and include cash-flow diagrams and examples.
2018	Sec A 3(c)	10	7	Critically explain why a well-known firm may submit a potential bid with zero profit.	Partial	10-11	5, 7	General bidding uncertainty and strategic decision support are covered, but the zero-profit rationale itself is not explicitly developed.

Examinations 2019-2021

Year	Original	Marks	QB p.	Question / task	Match	Primary	Support	Coverage assessment
2019	Sec B 6(a)	10	10	Describe the construction project development process from the contractor's perspective.	Exact	5-6	4	The wording closely matches the headings and process diagrams on Slides 5-6.
2019	Sec B 6(c)	13	10	Describe bid unbalancing and front payment as parts of a bidding strategy.	Exact	8-9	-	Both requested strategies are directly illustrated in Slides 8-9.
2019	Sec B 7(b)	10	10	Calculate the average bid price from field-supervision cost, total/basic cost and minimum/maximum head-office overhead, assuming nil profit.	Partial	7	5	Slide 7 supplies the tender-price/markup equation, but the special average-overhead calculation is not worked in this deck.
2020	Sec B 4(b)	18	13	Discuss the suitability of cost-reimbursement contracts and describe the construction project development process from the contractor's perspective.	Partial	5-6	-	Slides 5-6 fully support the development-process half. Cost-reimbursement suitability belongs to Lecture 02 - Construction Contracts.
2020	Sec B 5(a)	10	13	Define bid unbalancing and elaborate bid unbalancing and front payment as bidding strategies.	Exact	8-9	-	Slides 8-9 directly address the definition, cash-flow effect and examples of both strategies.
2020	Sec B 5(c)	10	13	Calculate the average bid price using field-supervision cost, total/basic cost and a 2%-4% head-office-overhead range, with nil profit.	Partial	7	5	The basic construction-bidding formula is present, but no worked average-overhead example is supplied.
2021	Sec B 6(b)	12	16	Calculate the average bid price from field-supervision cost and a 2%-4% office-overhead range, assuming nil profit.	Partial	7	5	Slide 7 gives tender price = estimated project cost + markup; the exam's averaging procedure requires a separate worked solution.
2021	Sec B 7(b)	10	16	Briefly describe the construction project development process from the contractor's perspective.	Exact	5-6	4	Slides 5-6 provide the requested contractor-perspective flow.
2021	Sec B 7(c)	15	17	Define bid unbalancing and describe bid unbalancing and front payment as parts of bidding strategy.	Exact	8-9	-	The question is a direct match to Slides 8-9.

Examinations 2022-2024

Year	Original	Marks	QB p.	Question / task	Match	Primary	Support	Coverage assessment
2022	Sec B 6(b)	15	19	Define bid unbalancing and briefly explain bid unbalancing and front payment as bidding strategies.	Exact	8-9	-	Slides 8-9 cover the complete requested topic with graphs.
2022	Sec B 6(c)	10	19	A managing director of a well-known firm chooses a zero-profit bid. Critically explain the rationale.	Partial	10-11	5, 7	The lecture supports the competitive-decision context but does not explicitly list the zero-profit motives.
2022	Sec B 7(b)	11	19	Define construction project development process and illustrate the typical process from the contractor's perspective.	Exact	5-6	4	The two contractor-perspective diagrams cover bid preparation, planning, execution, controls and completion.
2022	Sec B 7(c)	14	19	Use Alpha Private Limited's 62-bid history to draw a histogram and probability curve and determine the winning probability at 7.5% markup.	Exact	12-16	10-11	The lecture homework uses the same firm, same data and same 7.5% target; Slides 12-15 give the required method.
2023	Sec B 6(c)	10	21	Critically explain the rationale for a well-known contractor to submit a zero-profit bid.	Partial	10-11	5, 7	The deck explains competitive uncertainty and model limitations, but the specific rationale must be taken from class discussion or another source.
2023	Sec B 7(c)	15	21-22	Using the 62-bid Alpha data, construct the histogram/probability curve and find the chance of winning at 10% markup.	Direct	12-16	10-11	The dataset and method are the same as Slides 12-16, but the exam changes the target markup from 7.5% to 10%, so the final probability must be re-read from the new threshold.
2023	Sec B 8(a)	10	22	Describe bid unbalancing and front payment as bidding strategies, with examples.	Exact	8-9	-	Slides 8-9 contain the requested diagrams and examples.
2024	Sec B 7(c)	13	25	A schedule-of-rates contractor prices early activities high and later activities low; actual quantities then change. Identify the strategy and discuss cash flow, risk and final payment with a graph and example.	Direct	8	9	This is bid unbalancing/front loading. Slide 8 shows the income-versus-expenditure effect and gives high-early/low-late item examples. The detailed quantity-risk and final-payment discussion must be expanded from contract principles.
2024	Sec B 8(c)	17	25	Using internal cost, preliminary estimate, markup parameters, actual cost and change orders, compare several contract types and identify the least-beneficial contract.	Partial	7	5, 10-11	Slides 5 and 7 support the price/markup setup, while Slides 10-11 support bidding judgment. Full contract-by-contract numerical comparison requires Lectures 02 and 03.

3. Repeated Question Patterns

These clusters show how the same concept moves between Section A and Section B and how recent exams increasingly use scenarios rather than direct definitions.

#	Pattern	Past questions	Slides	Study implication
1	Bid unbalancing + front payment	2018 Q3(b); 2019 Q6(c); 2020 Q5(a); 2021 Q7(c); 2022 Q6(b); 2023 Q8(a); 2024 Q7(c)	8-9	7 appearances. Learn the graph, cash-flow purpose, early/late item pricing and the risk when actual quantities differ.
2	Construction project development - contractor perspective	2016 Q1(b); 2019 Q6(a); 2020 Q4(b); 2021 Q7(b); 2022 Q7(b)	5-6	5 appearances. Reproduce the two diagrams in sequence: bid preparation, then planning/execution/control/completion.
3	Zero-profit bid rationale	2017 Q8(a); 2018 Q3(c); 2022 Q6(c); 2023 Q6(c)	10-11 (partial)	4 appearances. The lecture supplies competitive context but not the full reasons; prepare a separate theory note.
4	Competitor-bid histogram and probability curve	2017 Q7(d); 2022 Q7(c); 2023 Q7(c)	12-16	3 appearances. The 2023 paper changes only the target markup to 10%.
5	Pricing structure / average bid price	2017 Q7(a); 2019 Q7(b); 2020 Q5(c); 2021 Q6(b); 2024 Q8(c)	5, 7	The basic formula is present; average-overhead and multi-contract calculations require other notes.
6	Typical tendering steps	2017 Q7(b)	4	Only one direct appearance, but it is an easy short question and the nine-step diagram is highly memorizable.

Year coverage

Year	Links	Marks	Main coverage
2016	1	8	Project development process
2017	4	30	Cost structure; tendering; Alpha probability; zero-profit rationale
2018	2	25	Bid unbalancing/front payment; zero-profit rationale
2019	3	33	Development process; bid unbalancing; average bid price
2020	3	38	Mixed contract/development question; bidding strategies; average bid price
2021	3	37	Average bid price; development process; bidding strategies
2022	4	50	Bidding strategies; zero-profit; development; Alpha probability
2023	3	35	Zero-profit; probability at 10%; bidding strategies
2024	2	30	Scenario-based unbalancing; multi-contract bid comparison

4. Slide-to-Question Reverse Index

Use this page while studying the deck: each slide block shows which examination questions it can answer or support.

Slides	Content	Mapped questions	Priority
3	Accuracy-in-bidding illustration	No direct standalone question identified	Low
4	Nine-step tendering procedure	2017 Q7(b)	Medium
5-6	Contractor project-development process; bid study, estimates, planning, execution and controls	2016 Q1(b); 2019 Q6(a); 2020 Q4(b); 2021 Q7(b); 2022 Q7(b)	Very high
7	Tender price = estimated cost + markup; percentage form and example	2017 Q7(a); 2019 Q7(b); 2020 Q5(c); 2021 Q6(b); 2024 Q8(c)	Very high
8	Bid unbalancing/front loading and cash-flow effect	2018 Q3(b); 2019 Q6(c); 2020 Q5(a); 2021 Q7(c); 2022 Q6(b); 2023 Q8(a); 2024 Q7(c)	Critical
9	Front/mobilization payment and income profile	Same 2018-2024 bidding-strategy cluster	Critical
10-11	Bid uncertainty, quality weighting and practical limits of models	Zero-profit questions in 2017, 2018, 2022 and 2023 (partial support)	High support
12-15	Competitor identification, markup formula, histogram and probability conversion	2017 Q7(d); 2022 Q7(c); 2023 Q7(c)	Critical
16	Alpha Private Limited 62-bid homework dataset	2017 Q7(d); 2022 Q7(c); 2023 Q7(c)	Critical
17	Thank-you slide	No question	None

Core formulas and recognition cues

$$\text{Tender Price} / \text{Project Value} = \text{Estimated Project Cost} + \text{Markup}$$

$$\text{Tender Price} = \text{Estimated Project Cost} \times (1 + \text{Markup \%})$$

$$\text{Competitor's Markup (\%)} = [(\text{Competitor's Bid} - \text{Your Estimated Cost}) / \text{Your Estimated Cost}] \times 100\%$$

High unit rates for early work + low rates for later work = bid unbalancing/front loading. A separate early mobilization receipt = front payment.

5. Exam-focused Study Plan

Recommended order for revising this lecture before the term final:

Order	Slides	Action
Step 1	Slides 8-9	Draw both cash-flow diagrams from memory. Explain why early items are overpriced, later items are underpriced, and how quantity changes can alter profit and risk.
Step 2	Slides 5-6	Memorize the two-stage contractor development flow: advertisement/bid study/estimating/submission, followed by planning/scheduling/resources/controls/completion.
Step 3	Slides 12-16	Practice computing competitor markup, grouping frequencies, drawing the histogram, converting to cumulative probability and reading the chance of beating the competitor.
Step 4	Slides 5 and 7	Memorize $\text{Price} = \text{Cost} + \text{Profit}$ and both tender-price formulas. Be able to classify cost, overhead and markup.
Step 5	Slide 4	Learn all nine tendering stages in order as an easy short-answer question.
Step 6	Slides 10-11	Prepare additional notes on when a contractor may accept zero profit, because the lecture gives context but not a complete exam answer.

Important limitations of this lecture

Not fully covered here	What to use instead
Detailed reasons for submitting a bid with zero profit	Use the instructor's discussion/class notes. Slides 10-11 only establish the competitive and decision-support context.
Worked average-bid-price calculation using minimum and maximum office-overhead percentages	Lecture 03 provides the cost structure; an instructor solution is still needed for the averaging convention.
Cost-reimbursement suitability and contract-type comparison	Lecture 02 - Construction Contracts.
Full 2024 Q7(c) analysis of quantity risk and final payment under Schedule of Rates	Combine Slide 8 with the Unit Price/Schedule of Rates material from Lecture 02.
Full 2024 Q8(c) numerical comparison among six contract types	Combine Lectures 02, 03 and 04; the present deck supplies only the pricing/markup foundation.
Completed solution to the Alpha homework table	Slides 12-15 explain the method and Slide 16 gives the data, but the final histogram/probability answer is not displayed.

Bottom line

For maximum return, prioritize Slides 8-9, 5-6, 12-16 and 7. These blocks account for the overwhelming majority of the 25 mapped question links and include the most frequently repeated and highest-mark questions.